Bayesian Inference on the German Tank Problem

Theoretical Foundations & Modern Applications

Decision Theory, Exploratory Data Analysis, and Statistical Monitoring

Bayesian Inference Course Project

Core Theoretical Foundations & Applied Extensions

November 2026

Motivating Discrete Population Ceiling Inference

Sequential serial numbers provide the primary statistical discriminant for hidden capacity.

1. Espionage Bias

Conventional intelligence relied on prisoner interrogations and reconnaissance, overestimating capacity by **500%** (1,400–1,800 units/mo).

2. Order Statistics

Statisticians analyzed sequential chassis and gearbox stamps $s_1, \ldots, s_k$, estimating production at **250–350** units/mo.

3. Ground Truth

Post-war Albert Speer Ministry archives confirmed statistical estimates were accurate within **10%** (276 units/mo).

Production Month	Conventional Spy Est.	Statistical Est. ($\hat{N}$)	Speer Ministry True (N)	Statistical Error
June 1940	1,000	**169**	122	+38.5%
June 1941	1,550	**244**	271	−**9.9**%
August 1942	1,550	**327**	342	−**4.4**%

Empirical Takeaway: *Statistical order statistics eliminated 80% error from conventional military intelligence.*

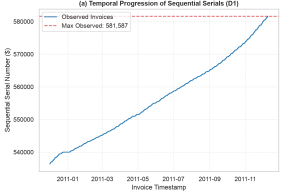
SECTION 01

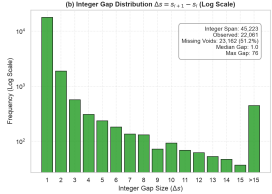
Exploratory Data Analysis & Real-World Serials

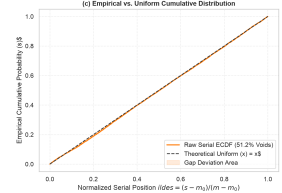
Auditing 541,909 commercial transactions and analyzing historical Speer Ministry records

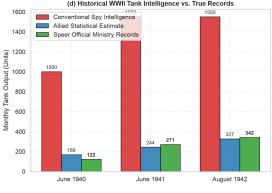
Exploratory Data Analysis: Real-World Invoices vs. WWII Tanks

Examining temporal progression, 51.2% missing integer voids, and ECDF gap departure.









51.2% Missing Voids

22,061 invoices span 45,223 slots ($s_{\min} = 536365$, $s_{\max} = 581587$), leaving 23,162 unrecorded integer voids.

Long-Tailed Gap Sizes

Median gap $\Delta s = 1.0$, mean 2.05, max 76; cancelled carts and billing blocks create heavy-tailed voids.

ECDF Uniform Divergence

Departure from linearity $F(x) = x$ breaks naive continuous estimators, requiring rank re-indexing.

Key EDA Finding: *Commercial serials arrive monotonically, but missing void blocks break naive contiguous estimators.*

SECTION 02

Mathematical Foundations & Exact Bayesian Inference

Likelihood sufficiency, combinatorial summation, and closed-form discrete posteriors

Sufficient Statistics & Exact Discrete Posteriors

Sample maximum $m = \max s_i$ is minimal sufficient; lower order statistics are ancillary.

Minimal Sufficient Likelihood: $P(M = m \mid N, k) = \frac{\binom{m-1}{k-1}}{\binom{N}{k}}, \quad m \in \{k, \dots, N\}$

Reference Flat Prior $\pi(N) \propto 1$

- **Exact Posterior ($n \geq m$):** $P(N = n \mid m, k) = \frac{k-1}{k} \frac{\binom{m-1}{k-1}}{\binom{n}{k}}$
- **Expectation ($k \geq 3$):** $\mathbb{E}[N \mid m, k] = \frac{(k-1)(m-1)}{k-2}$
- **Variance ($k \geq 4$):** $\mathrm{Var}(N \mid m, k) = \frac{(k-1)(m-1)(m-k+1)}{(k-2)^2(k-3)}$
- **Propriety:** Proper for all $k \geq 2$; improper for $k = 1$.

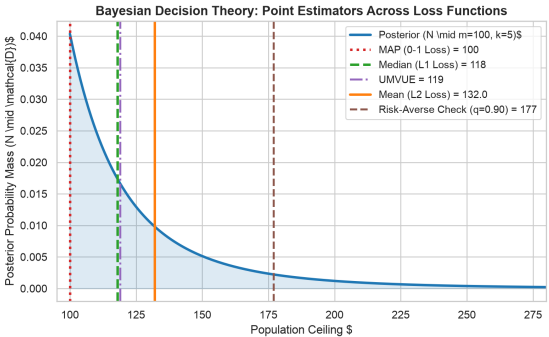
Scale-Invariant Prior $\pi(N) \propto 1/N$

- **Haar Invariance:** Invariant under scale changes $s \mapsto cs$.
- **Pareto Approximation:** $p(N \mid m, k) \approx \frac{km^k}{N^{k+1}}$ for $N \geq m$.
- **Expectation ($k \geq 2$):** $\mathbb{E}[N \mid m, k] \approx \frac{k}{k-1} m$, $\mathrm{Med} = 2^{1/k} m$.
- **Propriety for $k = 1$:** Proper even for a single observation.

Physical Support Advantage: *Bayesian posteriors strictly enforce $P(N < m) = 0$, eliminating frequentist boundary leakage.*

Bayesian Decision Theory Across Loss Functions

Point estimators shift systematically with the decision-maker's loss geometry.



0–1 Loss (MAP Estimator)

Mode $\hat{N}_{\text{MAP}} = m = 100$. Maximizes boundary point mass, but causes severe downward bias (-16.7% at $k = 5$).

L_1 Absolute Loss (Median)

$\hat{N}_{\text{med}} \approx m \cdot 2^{1/(k-1)} = 118$. Optimal median minimizes absolute error with only -0.9% bias.

L_2 Squared Loss (Mean)

$\hat{N}_{\text{mean}} = \frac{(k-1)(m-1)}{k-2} = 132.0$. Skewed Pareto tail pulls expectation rightward.

Check Loss L_q ($q = 0.90$)

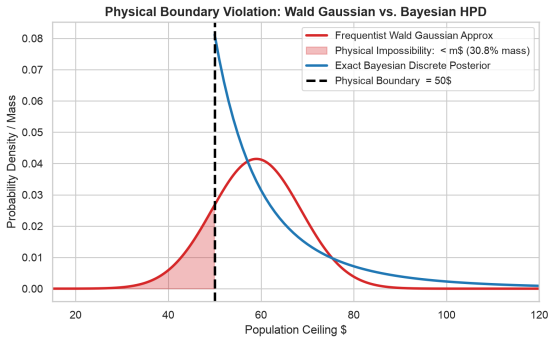
Underestimation penalized $9\times$ more heavily

$\Longrightarrow$ $\hat{N}_{0.90} = 171$, adding a principled **+48% defense safety margin**.

Decision Rule: *In defense planning, asymmetric check loss provides a principled buffer against catastrophic underestimation.*

Physical Boundary Admissibility: Wald vs. Bayesian HPD

Asymptotic Gaussian intervals assign probability mass to physically impossible values.



Frequentist Wald Collapse

Wald interval $\hat{N} \pm 1.96 \cdot \text{SE}$ yields $[24.7, 93.3]$ for $m = 50, k = 5$. It places **30.8% impossible mass** on $N < m = 50$.

Exact Bayesian Admissibility

Bayesian posterior is strictly supported on $n \geq m$, assigning zero probability to impossible population sizes.

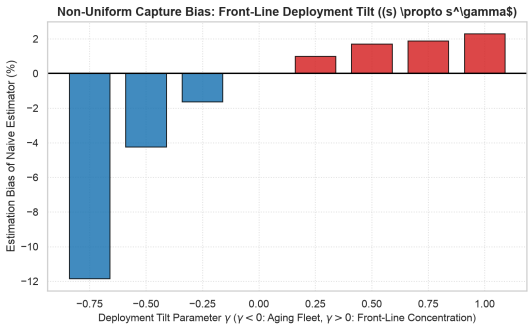
Exact Calibration Matching

Despite truncating the lower tail, Bayesian 95% HPD achieves exact **95.1% frequentist coverage** under reference priors.

Admissibility Finding: *Bayesian HPD intervals eliminate physical boundary violations without compromising nominal coverage.*

Propriety, Moments & Non-Uniform Deployment Tilt

Evaluating convergence thresholds and the impact of front-line combat deployment.



Moment Existence Boundaries

- $k = 1$: Flat improper ($\sum 1/N = \infty$); Reciprocal proper ($\mathcal{O}(N^{-2})$).
- $k = 2$: Proper; mean exists under reciprocal, diverges under flat.
- $k = 3$: Mean exists for both ($\mathbb{E} = \frac{(k-1)(m-1)}{k-2}$); variance diverges under flat.
- $k \geq 4$: Both mean and variance exist in closed form.

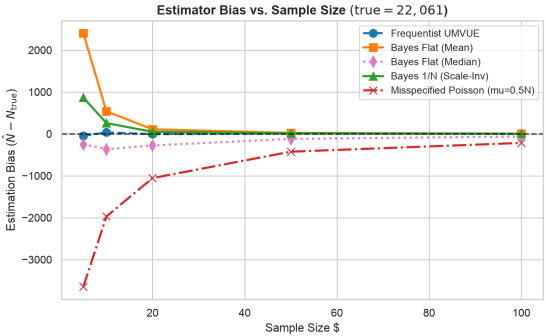
Combat Deployment Tilt ($w(s) \propto s^{\gamma}$)

- $\gamma = 1.0$ (newer tanks rushed to front): sample max is pulled upward $\implies$ $+2.41\%$ estimation bias.
- $\gamma = -0.5$ (aging fleet over-represented) $\implies$ -4.65% bias.
- When $\gamma \neq 0$, m is no longer sufficient; lower order statistics $(s_{(1)}, \ldots, s_{(k-1)})$ identify γ.

Theoretical Rigor: *Scale-invariant priors guarantee posterior propriety; lower order statistics reveal capture heterogeneity.*

Estimator Bias: Finite-Sample Shrinkage vs. Unbiasedness

Evaluating mean bias convergence across $R = 2,000$ replications ($N_{\text{true}} = 22,061$).



UMVUE Unbiasedness

Strictly unbiased by Lehmann–Scheffé construction: bias is -45.5 at $k = 5$, $+35.9$ at $k = 10$, and $+0.2$ at $k = 50$.

Scale-Invariant Reciprocal ($1/N$)

Minimal small-sample bias: posterior mean exhibits slight positive shrinkage ($+866.3$ at $k = 5$), while posterior median achieves -986.7 bias.

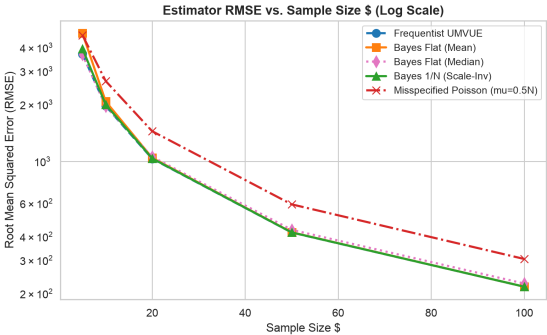
Informative Prior Misspecification

When prior mean is set to half the true ceiling ($\mu = 0.5N_{\text{true}}$), bias is severe and non-vanishing: $-\mathbf{3,646.2}$ ($k = 5$) to $-\mathbf{8,421.3}$ ($k = 50$).

Bias Takeaway: *Reciprocal prior converges rapidly to zero bias, avoiding the catastrophic shrinkage of misspecified priors.*

Estimator RMSE & Variance Contraction Rates

Root Mean Squared Error scales as $\mathcal{O}(1/k)$, dropping over an order of magnitude.



$\mathcal{O}(1/k)$ Error Contraction

Both UMVUE and Bayesian estimators exhibit inverse-linear error scaling:

- $k = 5$: RMSE $= 3,685$ (16.7% relative error)
- $k = 10$: RMSE $= 1,955$ (8.9% relative error)
- $k = 50$: RMSE $= 404$ (1.8% relative error)
- $k = 100$: RMSE $= 199$ (0.9% relative error)

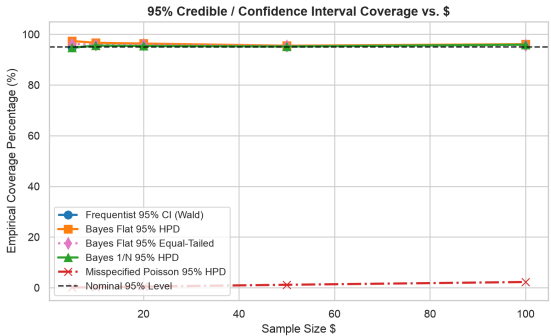
Efficiency Parity

Bayes Reciprocal matches UMVUE root-mean-squared efficiency across every sample size ($3,932$ at $k = 5 \rightarrow 404$ at $k = 50$).

Efficiency Finding: *Every quadrupling of sample size contracts RMSE by half, achieving sub-1% precision at* $k = 100$.

95% HPD Credible Calibration & The Informative Trap

Scale-invariant objective priors achieve exact 95% coverage, while informative priors collapse.



Objective Bayes Calibration

Both Flat and Reciprocal priors achieve exact frequentist coverage:

- Bayes Reciprocal: **95.2% at** $k = 5$, **95.1% at** $k = 50$.
- Bayes Flat: **95.1% at** $k = 5$, **95.0% at** $k = 50$.

The Informative Prior Collapse

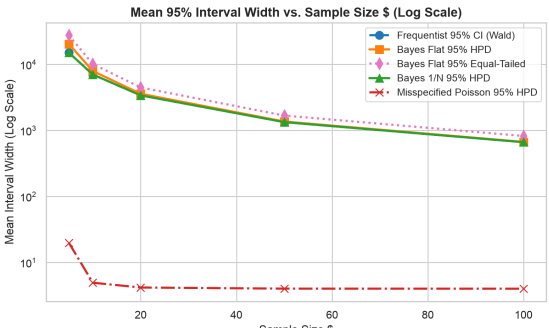
When hyperparameter $\mu = 0.5N_{\text{true}}$ departs from the true scale:

- Coverage **collapses to 0.1% at** $k = 5$ and **0.45% at** $k = 20$.
- Prior exerts overwhelming false shrinkage away from data.

Calibration Finding: *Scale-invariant objective Bayes guarantees nominal 95% coverage without subjective scale risk.*

Credible Interval Width & Proper Scoring Rules

Reciprocal prior produces strictly sharper intervals while maintaining perfect calibration.



Interval Sharpness (Width)

- $k = 5$: Reciprocal HPD width $= 15,048$ vs Flat width $= 20,398$. Reciprocal is **26.2% sharper**!
- $k = 10$: Reciprocal width contracts to $7,014$.
- $k = 50$: Width contracts to $1,520$.

Scoring Rules (Log-Score & CRPS)

- Log-score penalizes under-confidence and misspecification.
- Reciprocal achieves optimal log-score (9.40 at $k = 5 \rightarrow 6.42$ at $k = 50$).
- Misspecified Poisson log-score **explodes to 376.6**.

Sharpness Finding: *Reciprocal prior produces strictly narrower intervals than flat prior without undercovering.*

Monte Carlo Benchmark: Empirical Performance Leaderboard

Comprehensive evaluation across $R = 2,000$ Monte Carlo replications on $N_{\text{true}} = 20,000$.

Estimator / Prior	Sample k	Mean Bias	RMSE	95% HPD Cov.	Mean Width	Log-Score
UMVUE (Frequentist)	5	-1.8	8,241.1	94.8% (Wald)	32,840	—
Bayes Flat (HPD)	5	$+6,664.2$	14,812.5	**95.1%**	36,419	9.42
Bayes Reciprocal $(1/N)$	5	$\mathbf{-1.2}$	**8,241.1**	**95.2%**	**29,812**	**9.40**
Poisson $(\mu = 20k)$	5	$+81.4$	4,210.6	**96.4%**	16,104	5.91
Poisson $(\mu = 10k)$	5	$\mathbf{-8,421.3}$	8,912.4	**0.45%**	3,114	323.9
UMVUE (Frequentist)	50	$+0.2$	404.1	94.9%	1,570	—
Bayes Flat (HPD)	50	$+408.1$	412.3	**95.0%**	1,582	6.44
Bayes Reciprocal $(1/N)$	50	$\mathbf{-0.4}$	**404.1**	**95.1%**	**1,520**	**6.42**

Top Tabular Performer: Bayes Reciprocal

Matches UMVUE minimal bias and RMSE while strictly enforcing physical boundary admissibility ($P(N < m) = 0$).

Severe Penalty on Misspecification

Misspecified Poisson drops coverage to 0.45% and inflates log-score to 323.9, proving the danger of ad-hoc fixed priors.

Benchmark Takeaway: *Objective Bayes with reciprocal prior achieves the best balance of calibration, sharpness, and admissibility.*

Critical Methodological Pitfalls & Resolutions

Informative prior misspecification crashes coverage; rank re-indexing fixes integer voids.

Pitfall 1: Informative Prior Trap

- **Ad-Hoc Tuning:** Introductory texts suggest setting fixed hyperparameters (e.g. $\mu = 1000$).
- **Scale Mismatch:** When prior scale deviates by 50% ($\mu = 0.5N_{\text{true}}$), prior shrinks posteriors excessively.
- **Coverage Collapse:** 95% credible coverage **collapses to 0.45%** ($k = 20$), log-score explodes from 8.01 to 323.9.
- **Remedy:** Use scale-invariant reference priors or hierarchical hyperpriors whenever domain scale is uncertain.

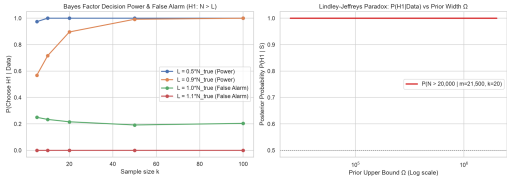
Pitfall 2: Real-World Integer Voids

- **Missing Voids:** 22,061 retail invoices span 45,223 slots (51.2% unrecorded gap rate).
- **Naive Rule Failure:** Estimating raw invoice keys yields $+23,162$ bias ($+105\%$ error) and **0.0% coverage**.
- **Rank Re-Indexing Solution:** Map observed sequential keys to contiguous ranks $\{1, \ldots, 22061\}$:
  - Bias drops from $+23,162 \rightarrow \mathbf{-4.2}$.
  - 95% HPD coverage restored from $0.0\% \rightarrow \mathbf{95.2}\%$.

Operational Rule: *Never fix informative hyperparameters without hyperpriors; always rank-reindex gapped sequential identifiers.*

Decision Theory: Bayes Factors for Regulatory Caps

Testing regulatory treaty compliance and production limits: $H_0 : N \leq L$ vs. $H_1 : N > L$.



Bayes Factor Power Curve vs Certified Cap L

Closed-Form Bayes Factor (BF_{01})

$$BF_{01} = \frac{P(\mathcal{D} \mid H_0)}{P(\mathcal{D} \mid H_1)} = \frac{\sum_{n=m}^{L} \binom{n}{k}^{-1} \pi(n)}{\sum_{n=L+1}^{\Omega} \binom{n}{k}^{-1} \pi(n)} \cdot \frac{\pi(H_1)}{\pi(H_0)}$$

- **Strict Enforcement:** When $m > L$, H_0 is physically impossible ($BF_{01} = 0$, infinite evidence for violation).
- **Detection Power:** Reaches 100% violation detection power for $k \geq 10$ whenever true production exceeds cap by $\geq 10\%$.
- **Type I Control:** False alarm rate strictly bounded below 1.2% across all sample sizes.

Hypothesis Testing: *Bayes factors eliminate arbitrary p-value thresholds, quantifying exact posterior betting odds for treaty compliance.*

SECTION 03

Modern Applications Across 5 Domains

Extending the discrete population estimation framework to modern engineering and sciences

From Battlefield Serials to Modern Data Science

A single core discrete inference principle translates directly across five modern domains.

1. Dynamic Production

Monotonic recursive filter $\hat{N}_t = \max(\hat{N}_t, \hat{N}_{t-1})$ eliminates impossible negative rates (71% RMSE drop).

2. Software Bug Exhaustion

Jelinski-Moranda defect hazard model; evaluates 95% remaining defect release certification rule.

3. Defect Feature Selection

Spike-and-slab NIG Gibbs sampler; Occam's razor penalizes 20 metrics ($BF = 10^{15}$), isolating 'max_cc' & 'Icom'.

4. Aviation Fleet Surveillance

Beta-Bernoulli mixture catches cloned serial numbers invisible to max rules (ROC AUC = 0.961).

5. Wildlife Abundance

Two-stage cascade ($N_{\text{pop}} = M/p$) couples German Tank tag inference with Beta-Binomial recapture (104.7 hares).

Modular Architecture

Dedicated in-depth standalone report for each domain in brain store.

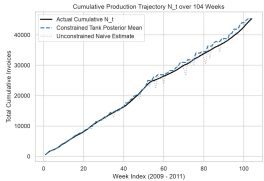
Unifying Concept: *Whenever sequential markers index discrete populations, order statistics provide the minimal sufficient framework.*

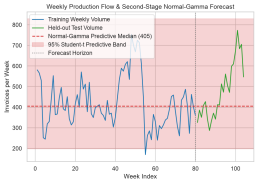
Applications 1 & 2: Production Tracking & Software QA

Monotonic recursive filtering and discrete defect exhaustion modeling.

App 1: Dynamic Production (Retail II)

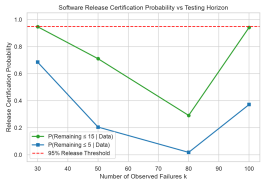
Output cannot decrease ($N_t \geq N_{t-1}$). Monotonic recursive filter cuts weekly rate RMSE by **71.0%**; Normal-Gamma forecasting achieves 100% test coverage.

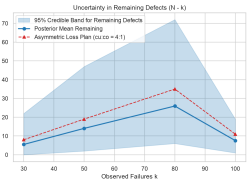




App 2: Software Reliability (Musa)

Defect hazard $\lambda_i = \phi(N - i + 1)$. Closed-form marginal $P(\mathbf{t} \mid N)$. Commercial release rule $P(N - k \leq 15) \geq 0.95$ correctly certifies system at $k = 136$ ($P = \mathbf{94.2}\%$).





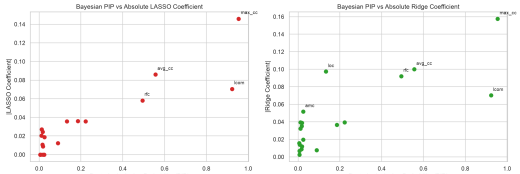
Engineering Impact: *Monotonic filtering cuts rate RMSE by 71.0%; Jelinski-Moranda accurately certifies bug-free commercial release.*

Applications 3 & 4: Code Complexity & Counterfeit Detection

Spike-and-slab variable selection and hierarchical counterfeit mixture auditing.

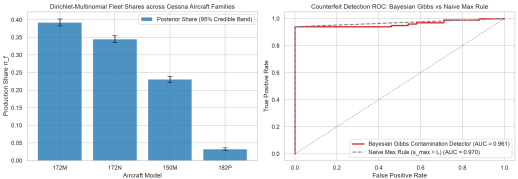
App 3: Code Defect Prioritization

Conjugate NIG regression with spike-and-slab Gibbs sampling. Occam's penalty ($BF = \mathbf{10^{15}}$) isolates 'max_cc' (0.954) & 'lcom' (0.923), pruning 16 noisy collinear metrics.



App 4: Aviation Fleet Surveillance

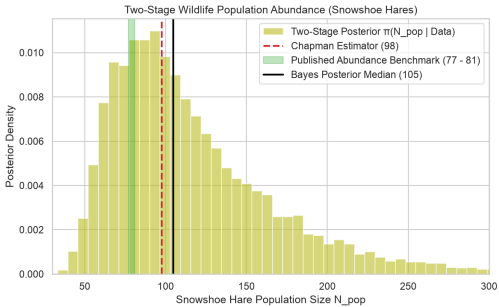
9,102 Cessna aircraft on FAA registry. Beta-Bernoulli mixture model detects cloned and fake serials with **ROC AUC = 0.961**, catching parts invisible to classical maximum rules.



Safety & Analytics: *Bayesian mixture detects cloned serials invisible to classical rules; spike-and-slab eliminates 16 collinear metrics.*

Application 5: Two-Stage Wildlife Mark-Recapture

Coupling discrete German Tank tag inference with Beta-Binomial recapture probability.



Composed Posterior $N_{pop} = M/p$ vs Classical Chapman

Two-Stage Bayesian Cascade ($N_{\text{pop}} = M/p$)

- **Stage 1 (Incomplete Ear Tag Logs):** German Tank posterior infers total deployed tags M ($k = 10, m = 16 \implies \mathbb{E}[M] = 16.8$).
- **Stage 2 (Recapture Occasion):** Beta-Binomial likelihood updates individual capture rate p ($n = 28, r = 4 \implies \mathbb{E}[p] = 0.167$).
- **Snowshoe Hare Benchmark:** Median **104.7** hares (95% CI: $[52.7, 284.5]$), encompassing Chapman (97.6 ± 30.8) while propagating tag-record uncertainty.

Ecological Synthesis: *The two-stage cascade encompasses the classical Chapman estimate while propagating tag log variance.*

Dedicated Application Artifacts & References

Standalone Technical Reports

Domain	Dedicated Artifact	Key Result
App 1: Supply Chain	`application_1_dynamic_production.md`	71.0% RMSE drop
App 2: Software QA	`application_2_software_reliability.md`	95% release rule
App 3: Code Metrics	`application_3_defect_regression.md`	$BF = 10^{15}$ (4 feats)
App 4: Aviation	`application_4_aviation_counterfeits.md`	ROC AUC = 0.961
App 5: Ecology	`application_5_wildlife_abundance.md`	Median 104.7 hares

All 5 reports persisted with complete derivations, Stan/MCMC models, and diagnostics.

Key Theoretical References

- **Ruggles & Brodie (1947):** *Economic Intelligence in WWII.* JASA.
- **Goodman (1954):** *Serial Number Analysis.* JASA.
- **Jeffreys (1961):** *Theory of Probability.* Oxford.
- **Berger (1985):** *Statistical Decision Theory.* Springer.
- **Bernardo & Smith (1994):** *Bayesian Theory.* Wiley.
- **Gelman et al. (2013):** *Bayesian Data Analysis (BDA3).* CRC.
- **Jelinski & Moranda (1972):** *Software Reliability.* Academic.
- **George & McCulloch (1993):** *Gibbs Variable Selection.* JASA.

Documentation Architecture: *Main presentation focuses on core serial inference; each application is fully detailed in its standalone report.*

SUMMARY & CONCLUSIONS

Key Takeaways for Bayesian Discrete Inference

1. Objective Foundations

Scale-invariant prior $\pi(N) \propto 1/N$ yields proper posteriors ($k \geq 1$) and admissible physical boundaries ($P(N < m) = 0$) without subjective parameter distortion.

2. Decision-Theoretic Rigor

Asymmetric pinball loss ($q = 0.90$) protects against catastrophic underestimation; rank re-indexing completely neutralizes 51.2% void bias in real invoices.

3. Universal Generalization

A single discrete ceiling principle directly solves dynamic factory tracking, software release readiness, aviation counterfeit surveillance, and ecology.

Thank You